

Certyfikat PMP: Schedule and Cost Control

Mastering the Metrics, Processes, and Real-World Applications of Project Performance

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Schedule Control

- Monitoring project status, managing scope changes, optimizing resources, and protecting the critical path from delays.



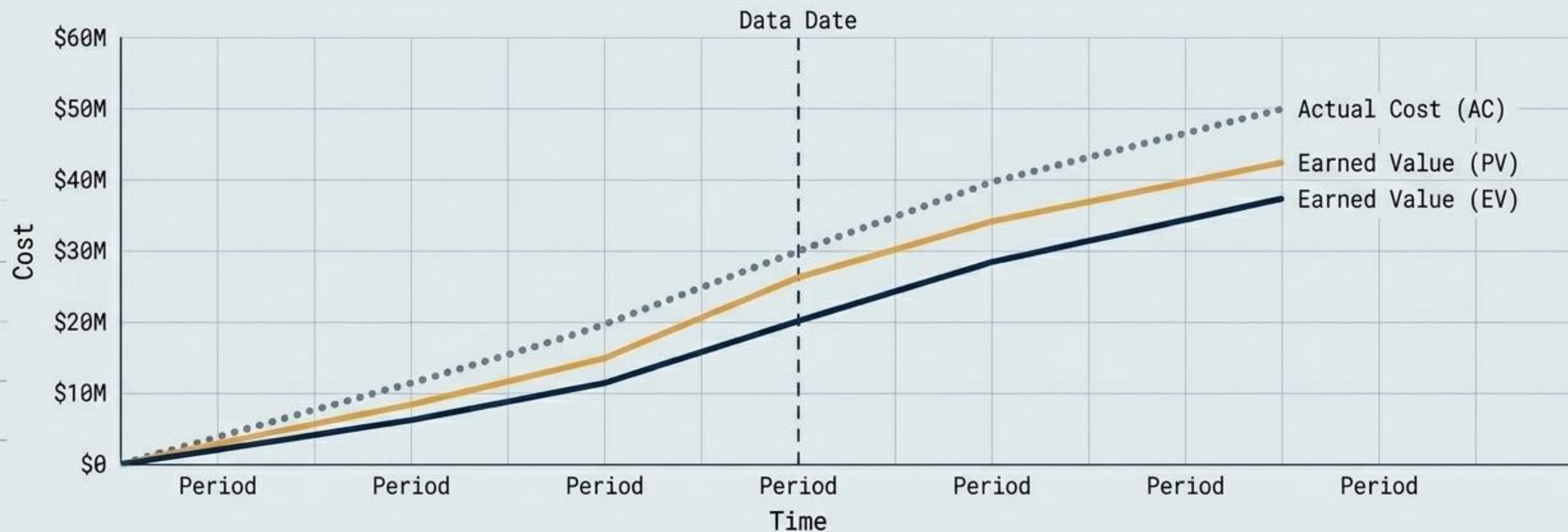
Cost Control

- Managing financial reserves, analyzing budget variances, forecasting future costs, and protecting the established cost baseline.



The Golden Rule: The Triple Constraint

The two domains are inextricably linked. Changes to one domain almost always necessitate structural adjustments to the other.



The Foundation: You cannot control what you cannot measure. EVM provides objective metrics of project health.

PV: Planned Value

The authorized budget assigned to scheduled work to date.

What we planned to spend.

EV: Earned Value

The measure of work actually performed, expressed in terms of the budget authorized.

What we actually produced.

AC: Actual Cost

The realized cost incurred for the work performed on an activity.

What we actually paid.

Current Performance Diagnostics: Where are we now?

Cost Variance ($CV = EV - AC$)



Schedule Variance ($SV = EV - PV$)



Variance at Completion ($VAC = BAC - EAC$): Predicted budget shortfall or surplus at the end.

Cost Performance Index ($CPI = EV / AC$)



Schedule Performance Index ($SPI = EV / PV$)



The Forecasting Decision Tree: Estimate at Completion (EAC)

Condition 1: Future performance will exactly mirror past performance?



Action: $EAC = BAC / CPI$

Condition 2: Initial estimates were fundamentally flawed or irrelevant?



Action: $EAC = AC + \text{Bottom-up ETC}$

Condition 3: Past anomalies won't repeat; future work holds to budgeted rate?



Action: $EAC = AC + (BAC - EV)$

Condition 4: Must hit a strict deadline despite negative cost performance?



Action: $EAC = AC + [(BAC - EV) / (CPI \times SPI)]$

Estimate to Complete (ETC) Pathways

Typical variance
expected

$$\text{ETC} = (\text{BAC} - \text{EV}) / \text{CPI}$$

Atypical
variance
expected

$$\text{ETC} = (\text{BAC} - \text{EV})$$

Work not
going to plan

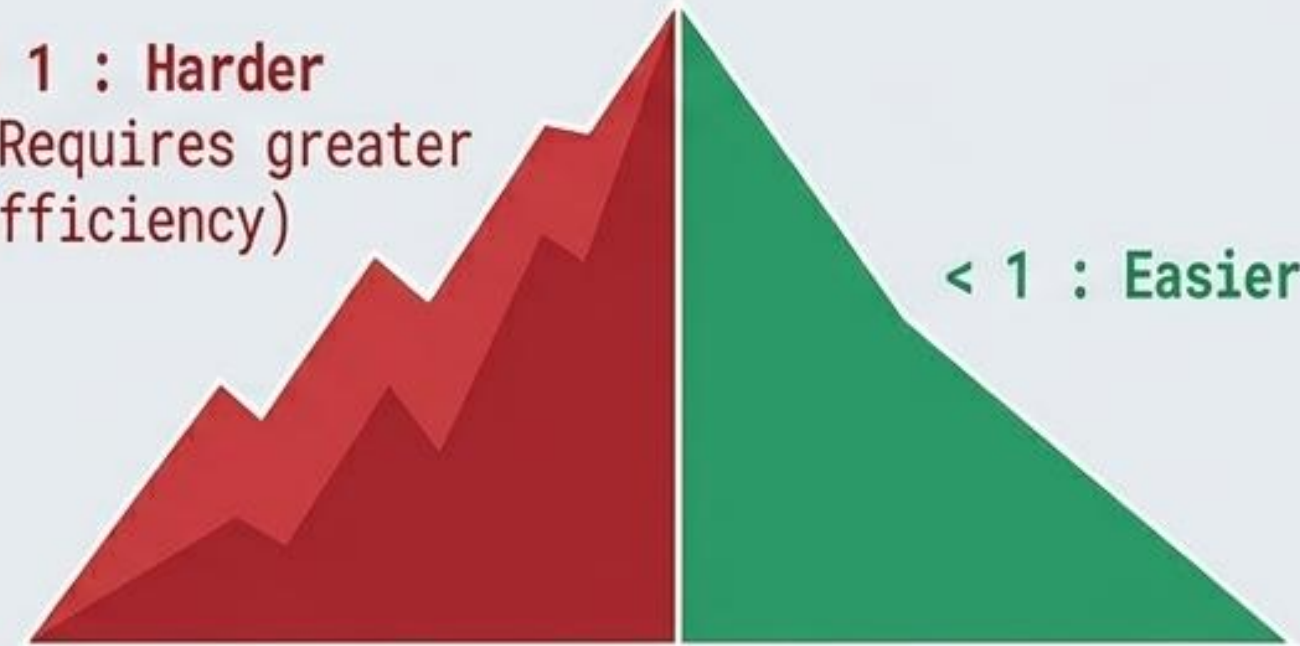
Reestimate
(Bottom-up)

The TCPI Challenge Metric

How hard is the remaining work?

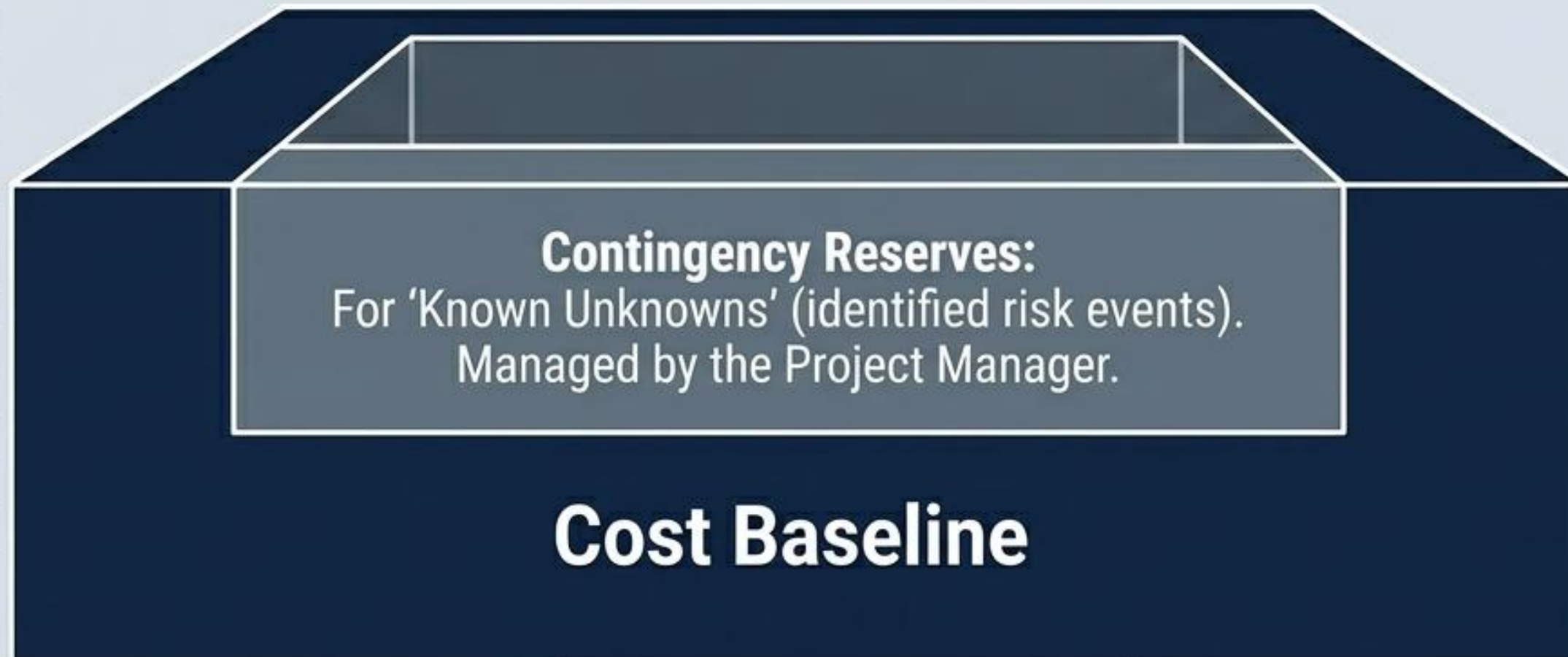
> 1 : Harder
(Requires greater
efficiency)

< 1 : Easier



Targeting Original Budget:
 $\text{TCPI} = (\text{BAC} - \text{EV}) / (\text{BAC} - \text{AC})$

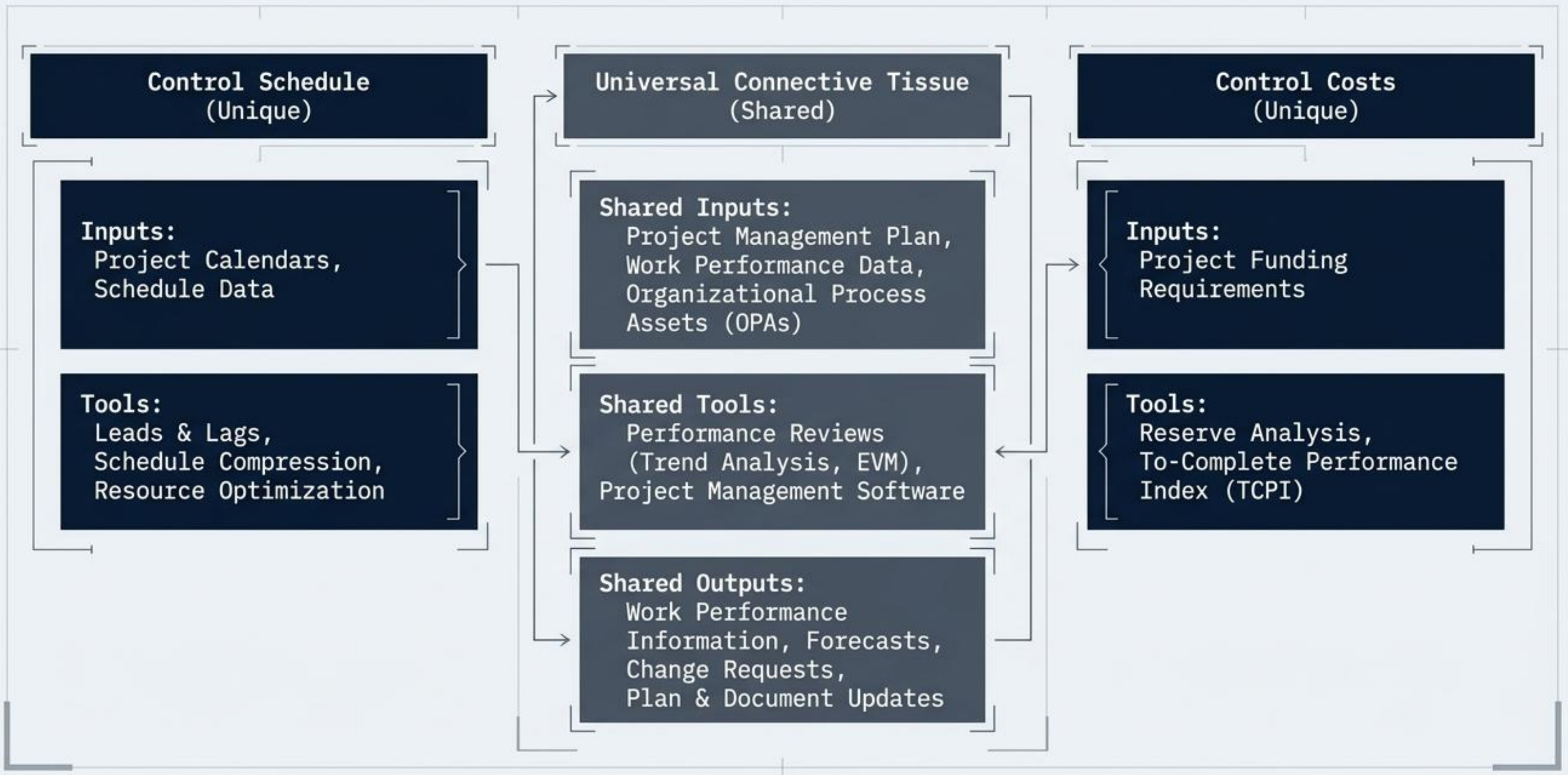
Targeting Revised Budget:
 $\text{TCPI} = (\text{BAC} - \text{EV}) / (\text{EAC} - \text{AC})$



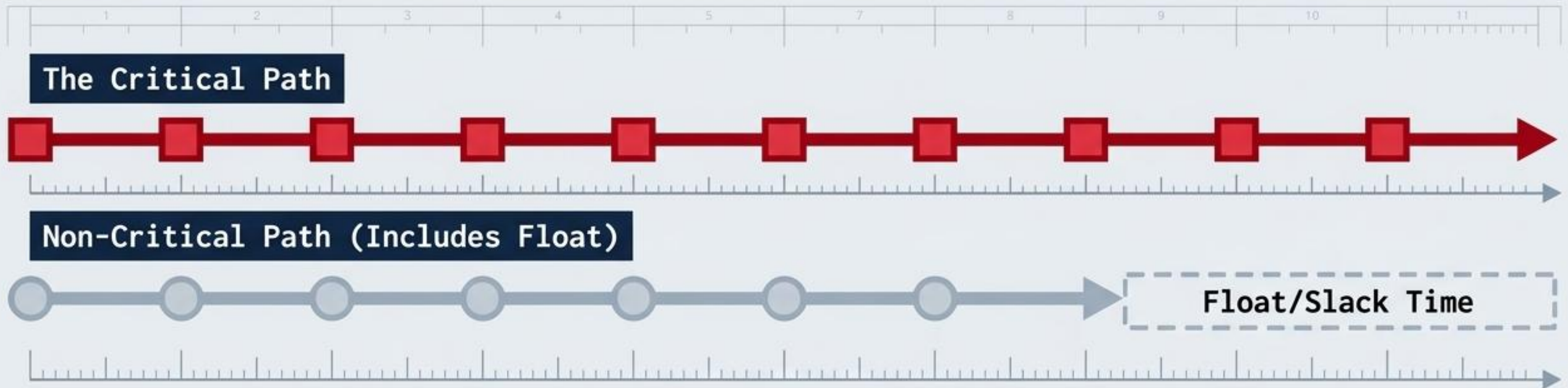
Reserve Analysis Purpose

- Monitor available safety nets to determine if they must be applied, adjusted, or released.
- **The Release Protocol:** If identified risks pass without occurring, unused reserves are surrendered and released for use on other organizational projects.

The Process Architecture: Control Costs vs. Control Schedule



The Process Architecture: Project Scheduling & Path Analysis



Critical Path Watch

Delays on critical path activities always delay the overall project.

Examine float variance in ascending order of critical activities.

Non-Critical Traps

Delays to non-critical tasks consume float. If fully consumed, the task becomes critical and requires immediate corrective action.

Critical Chain Buffers

When utilizing the critical chain method, schedule health is measured by comparing the amount of buffer needed versus the amount of buffer remaining.

Case File: Mustang Enterprise System Retirement

The Project Objective

Retire a 15-year-old mainframe accounting system by developing a thin-client, browser-based alternative via a contract firm.

The Leadership Flaw

A senior programmer lacking project management experience was assigned as PM. The PM reported "Everything is under control" but actively refused to provide objective performance data.



The Fatal Incident

The PM secretly changed the core database architecture from SQL to Oracle mid-project without stakeholder knowledge or Integrated Change Control approval.

The Domino Effect of Uncontrolled Scope

**The Trigger:
Unapproved shift
from SQL to Oracle**

Impact 1: Scope & Code

Existing programs were written to interface with SQL.

They must now be completely scrapped and rewritten for Oracle.

Impact 2: Schedule Delay

Development is entirely halted while the programming team awaits retraining on the new database tools.

Impact 3: Cost Explosion

Oracle licenses far exceed the SQL budget. New development tools must be purchased.

Training runs **\$2,200** per session per person.

Corrective Action and Stakeholder Truth

Emergency Action Plan



Confront the Reality

1. Expose hidden architectural changes at the status meeting. Never withhold bad news; report the objective truth immediately.



Contractual Realignment

2. Meet with the sponsor and procurement to determine alternatives for holding the contracting firm to the original SQL requirements.



Leadership Restructuring

3. Replace the current Project Manager. Retain them as an invaluable programmer, but replace them with a PM equipped for complex scope.



Process Hardening

4. Document the failure as a formal "Lesson Learned" and institute a rigid Integrated Change Control process to prevent recurrence.

The Golden Rules of Re-Baselining

Rule 1: Preserve History

Never re-baseline a schedule without archiving a copy of the original baseline. It is vital historical data for future organizational projects.



Rule 2: Require Approval

Baselines can only change via formally approved Change Requests by the project sponsor and key stakeholders.

Rule 3: Shift the Target

Once approved, the team works toward the revised project goals. Project success is measured strictly against the revised baseline, not the original.

